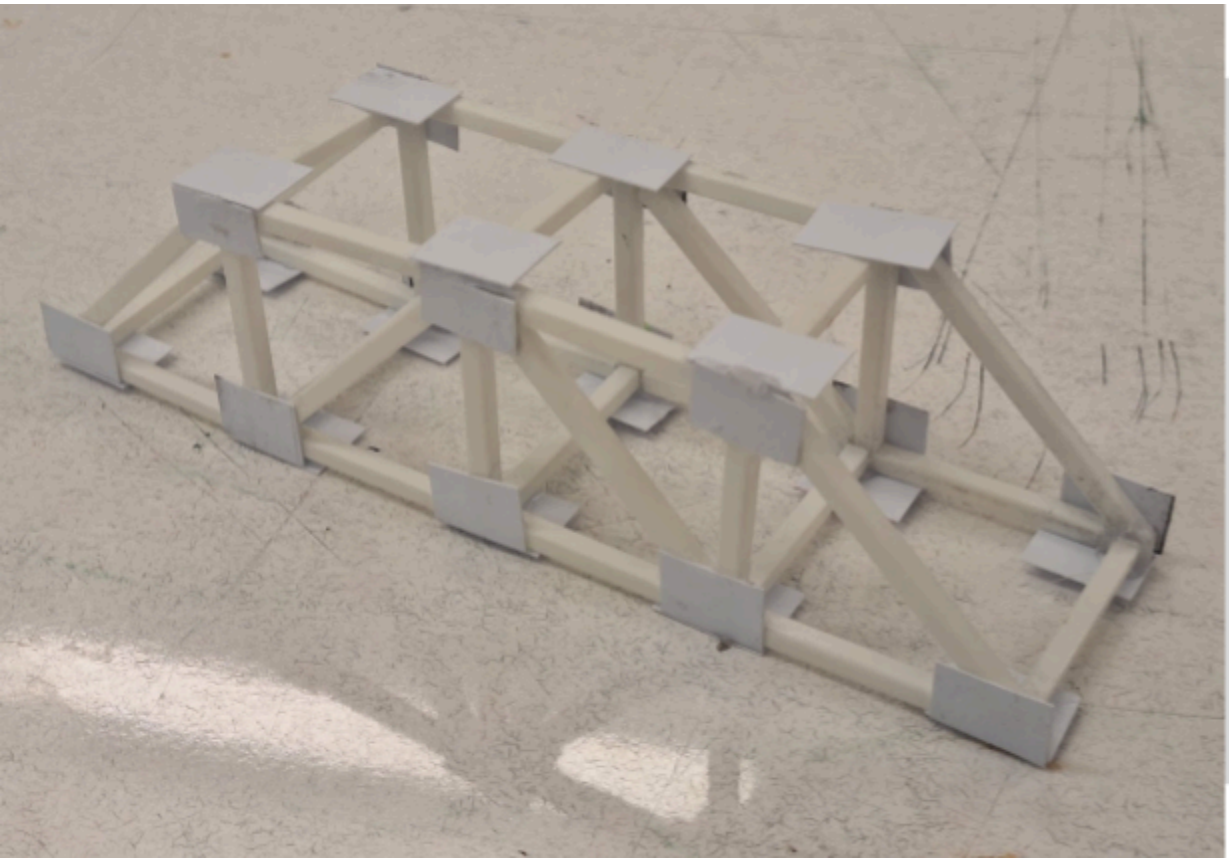
ME20 Truss Project

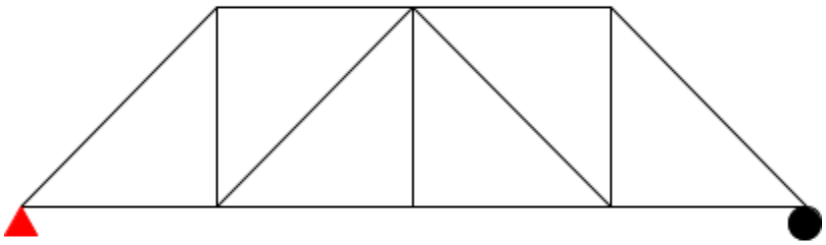
Breoghan Conchas Marquez and Raphael Lemus Solis



Howe Truss

Designing the Truss

M: Number of Joints
N: Number of Bars
R: Number of Support Reactions
2M = N + R

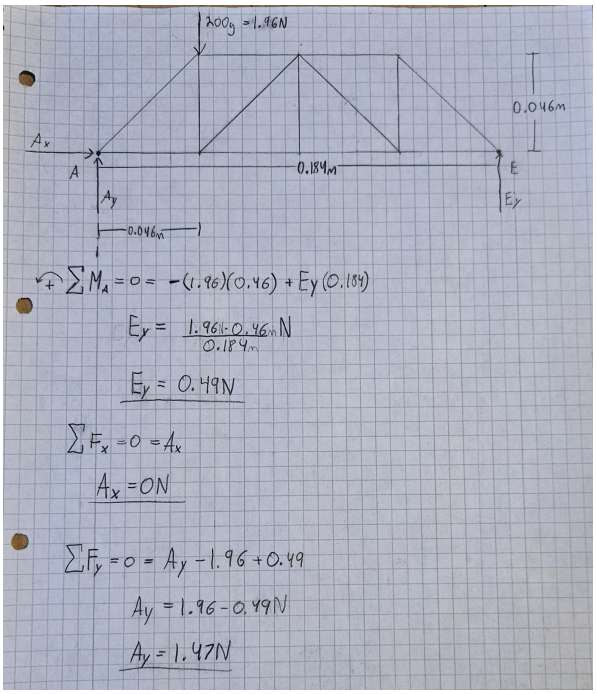


M = 8; N = 13; R = 3

Figure 1: The truss is statically determinate

Theoretical Calculations for Support Reactions

Loading Condition 1



Loading Condition 2

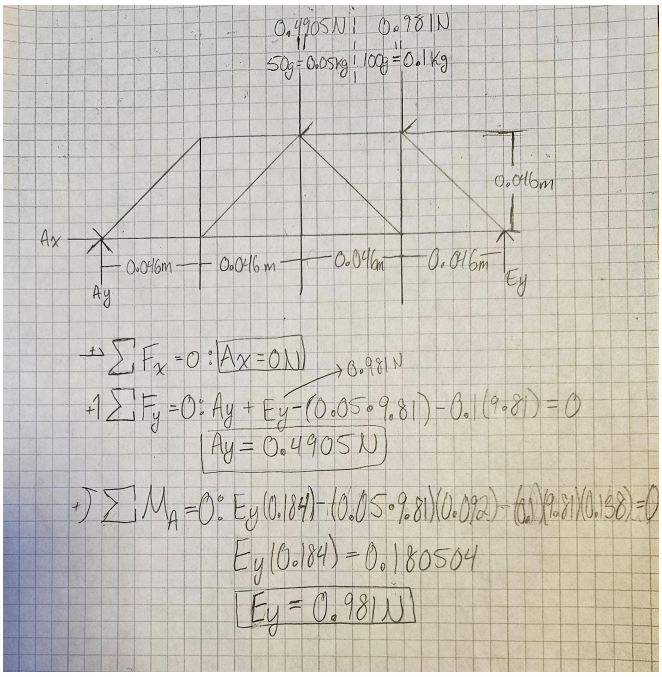
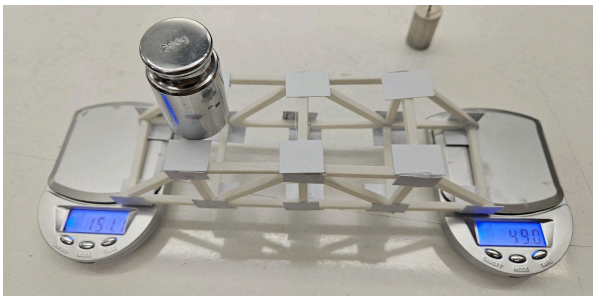


Figure 2: Free Body Diagram (FBD) of the truss under both loading conditions for analysis

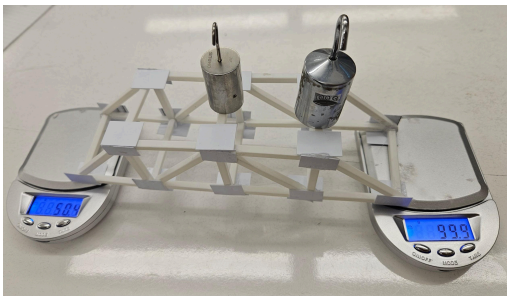
Building the Truss and Finding Support Reactions

Loading Condition 1



1.4808N ---------- 0.4802N

Loading Condition 2



0.490N ---------- 0.979N

Figure 3: ABC plastic truss and the support reactions for both loading conditions

Comparing Calculations to Measurements

		Pin		Roller
		A_x (N)	A_y (N)	E_y (N)
Condition 1	Theoretical	0	1.47	0.49
	Measurement	0	1.481	0.480
	Online Calculator	0	1.47	0.49
	Matlab	0	1.47	0.49
Condition 2	Theoretical	0	0.490	0.981
	Measurement	0	0.490	0.979
	Online Calculator	0	0.4905	0.981
	Matlab	0	0.49	0.981

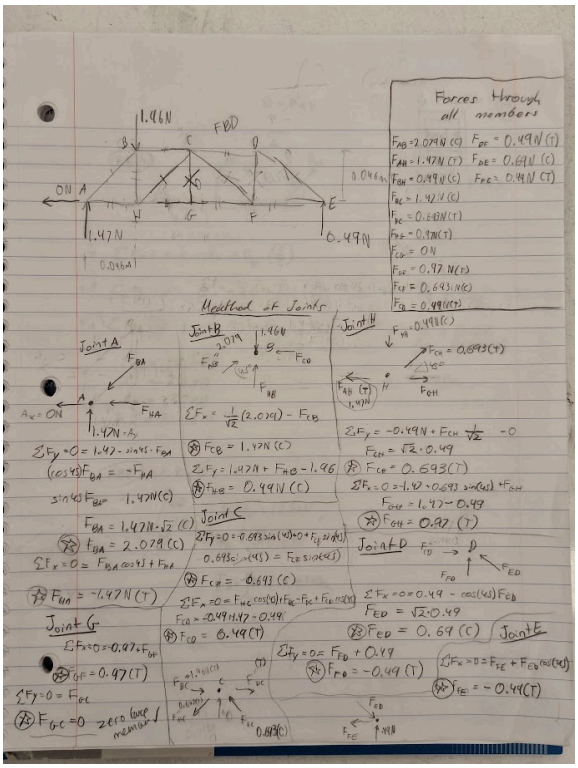
Figure 4: Calculations of the reaction forces for both loading conditions

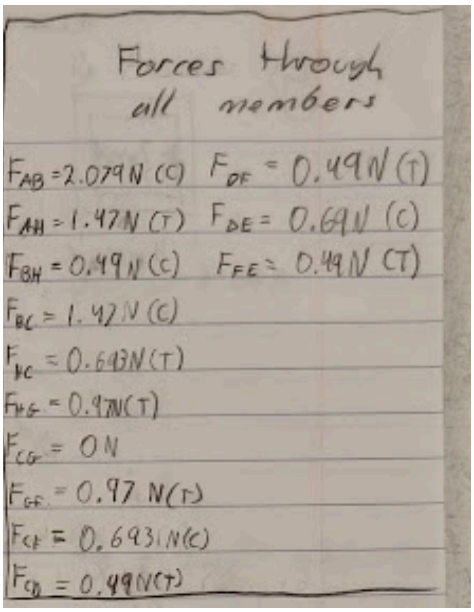
Calculating Members' Internal Loads

Assumptions made: Each angle was either a 45- or 90-degree angle & it is in equilibrium

Loading Condition 1

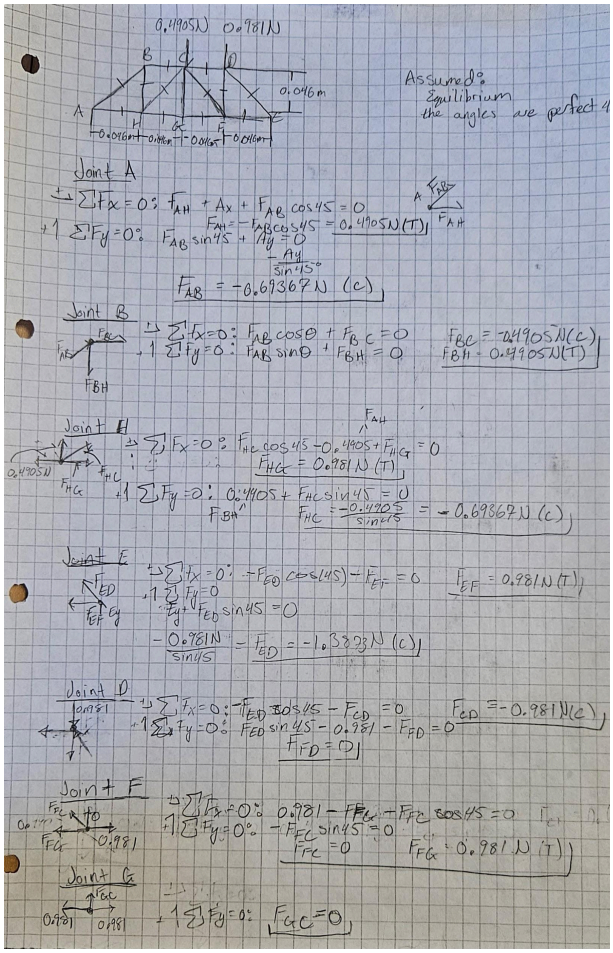
Method of Joints:





Loading Condition 2

Method of Joints:



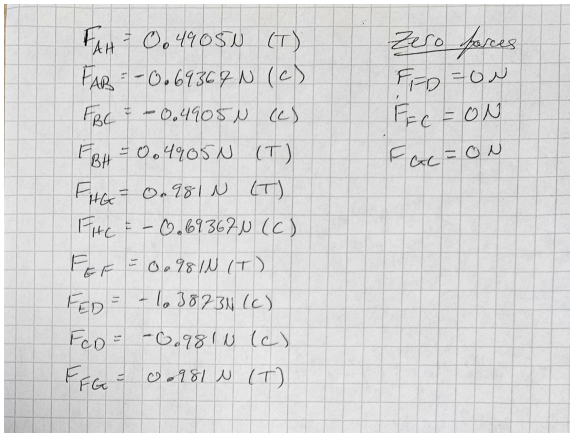


Figure 5: Theoretical calculations of each member for both loading conditions

Checking Answers With Online Truss Calculator

Loading Condition 1

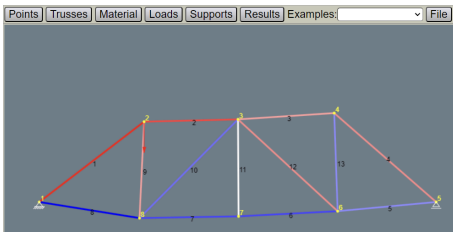
All correct 👍

Truss Forces

i	N_i	σ_i
1	-2.079	-2.970e+5
2	-1.470	-2.100e+5
3	-0.4900	-70000
4	-0.6930	-98995
5	0.4900	70000
6	0.9800	1.400e+5
7	0.9800	1.400e+5
8	1.470	2.100e+5
9	-0.4900	-70000
10	0.6930	98995
11	0.000	0.000
12	-0.6930	-98995
13	0.4900	70000

Reaction Forces

i	F_{ix}	F_{iy}
1	0.000	1.470
5	0.000	0.4900



Loading Condition 2

All correct 👍

Truss Forces

i	N_i	σ_i
1	-0.6937	-99096
2	0.4905	70071
3	0.4905	70071
4	-0.6937	-99096
5	-0.4905	-70071
6	0.000	0.000
7	0.9810	1.401e+5
8	0.9810	1.401e+5
9	0.000	0.000
10	-0.9810	-1.401e+5
11	0.000	0.000
12	0.9810	1.401e+5
13	-1.387	-1.982e+5

Reaction Forces

i	F_{ix}	F_{iy}
4	0.000	0.9810
8	0.000	0.4905

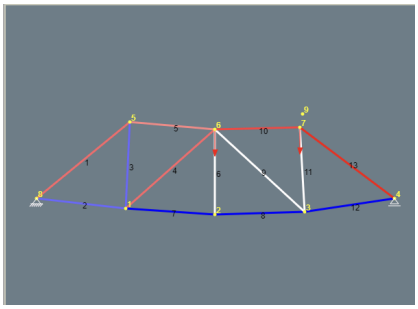


Figure 6: Online truss calculator calculations for both loading conditions

CONDITION 1	Theory		Online Truss Calculator		Error %
Members	Force (N)	T or C	Force (N)	T or C	
AH	1.47	T	1.47	T	0
AB	2.079	C	2.079	C	0
BC	1.47	C	1.47	C	0
BH	0.49	C	0.49	C	0
HG	0.98	T	0.98	T	0
HC	0.693	T	0.693	T	0
EF	0.49	T	0.49	T	0
ED	0.69	C	0.693	C	0
CD	0.49	C	0.49	C	0
FG	0.98	T	0.98	T	0
FD	0.49	T	0.49	T	0
FC	0.693	C	0.693	C	0
GC	0	-	0	-	0

Figure 7: A table of the forces on each member for condition 1

CONDITION 2	Theory		Online Truss Calculator		Error %
Members	Force (N)	T or C	Force (N)	T or C	
AH	0.4905	T	0.4905	T	0
AB	0.6937	C	0.6937	C	0
BC	0.4905	C	0.4905	C	0
BH	0.4905	T	0.4905	T	0
HG	0.981	T	0.981	T	0
HC	0.6937	C	0.6937	C	0
EF	0.981	T	0.981	T	0
ED	1.387	C	1.387	C	0
CD	0.981	C	0.981	C	0
FG	0.981	T	0.981	T	0
FD	0	-	0	-	0
FC	0	-	0	-	0
GC	0	-	0	-	0

Figure 8: A table of the forces on each member for condition 2

MATLAB Truss Analysis

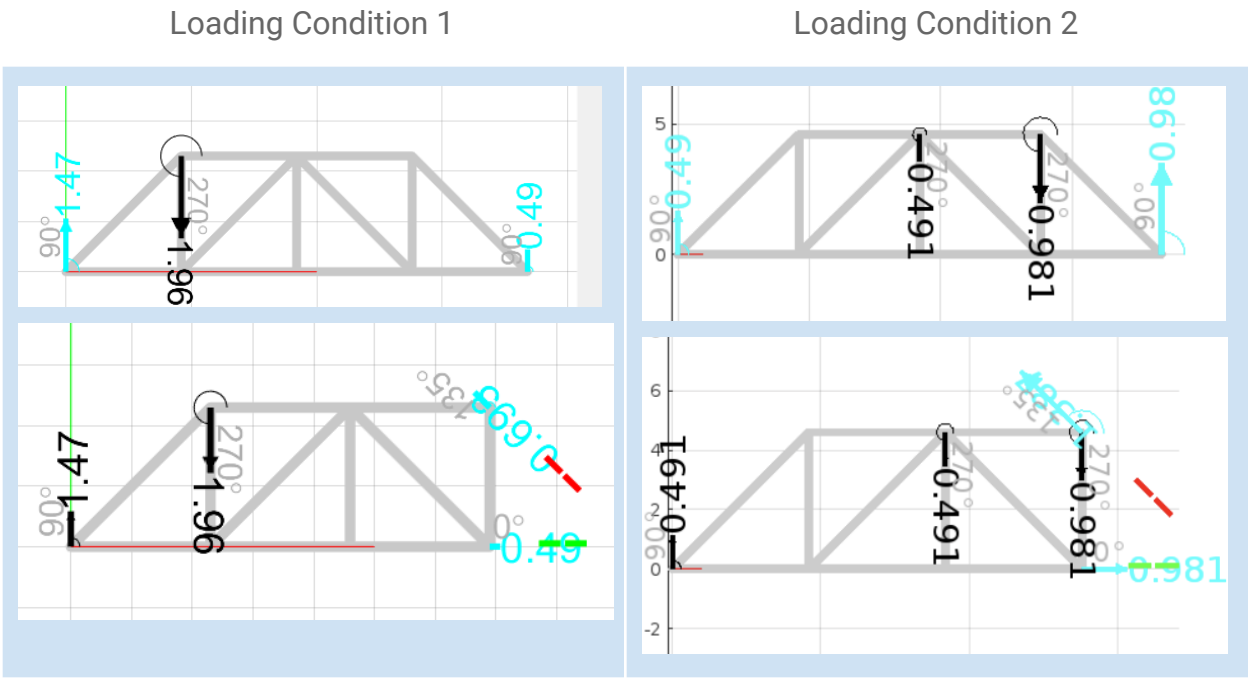


Figure 9: The two loading conditions with the MatLab code

CONDITION 1	Theory		MATLAB		Error %
Members	Force (N)	T or C	Force (N)	T or C	
EF	0.49N	T	0.49	T	0
ED	0.69N	C	0.693	C	0

Figure 10: Two members of condition 1

CONDITION 2	Theory		MATLAB		Error %
Members	Force (N)	T or C	Force (N)	T or C	
EF	0.981	T	0.981	T	0
ED	1.387	C	1.387	C	0

Figure 11: Two members of condition 1

Conclusion

Summary:

In the beginning, it was confirmed that the truss was statistically determinant due to the equation *2M = N + R.* Therefore, we can determine that the truss bridge can be in equilibrium and only use the equilibrium equations to find the values of the reaction forces. The reaction forces fell on the pin at A and the roller at B. The theoretical calculation for condition one was $A_x = 0$ *N,* $A_y = 1.47$ *N, and* $E_y = 0.49$ *N.* However, for the actual measurement of the truss bridge, $A_x = 0$ *N,* $A_y = 1.481$ *N, and* $E_y = 0.480$ *N.* The difference between A_y is 0.011 N, and the difference between E_y is 0.01 N.

On the other hand, condition two had a difference of 0 N for A_x and A_y and 0.002 N for E_y. Therefore, the measurement was relatively close to the theoretical calculations, which proves that the bridge was built accurately to the assumptions made. The assumption made throughout the project was that each angle was either 45- or 90-degree. Also, the lengths were accurately built according to the theoretical calculations.

According to the calculator, for the various members for each load, there was a 0% error, meaning that the theoretical calculations made for each member were accurate. The method of joints was used to calculate each member to ensure that the calculations were correct for each member. Therefore, this meant that there had to be a lot of joints to be taken into account to get each member. Another interesting fact about both loading conditions was that there were zero members. For condition one, the only zero member was GC. However, there were three zero members for condition two: FD, FC, and GC. Also, six members were in tension, and six were in compression for condition one. However, for condition two, five members were in tension, and five were in compression. A Matlab code was used to calculate two of the thirteen members to ensure the calculations for each member were correct. The members used for both loading conditions were EF and ED. Between the theoretical and MatLab calculations, there was a 0% error, giving the same tension and compression for the two members.

What We Learned:

The project taught us about the complexities of the bridge-building technique for trusses. Many assumptions were made before and after, surprising us due to measurements being relatively close to the theoretical calculations. Although the project had MatLab code to do, which we needed to become more experienced with, it was a stepping stone on how different websites and personal code ensure the safety of the design. In the future, we will double-check the calculations using various software to ensure the machine's or building's system does not fail and hinder the safety of the people. Not only that, but we also step inside the engineer’s mindset to figure out the load distribution of each beam. Therefore, we got to understand the beauty of triangles and how a truss bridge is considered a sturdy variant of a bridge. With this knowledge in mind, we could go into engineering to design efficient structures with the materials and forces in mind. The forces applied to the beams could tell the engineer

whether the structure is going past the material’s yield strength. Therefore, it was a way to connect both engineering classes: this class and Materials and Manufacturing I.